

```
Code
SELECT COUNT(DISTINCT job_id) AS Jobanzahl
FROM employees;

Ausgabe
```

	JOBANZAHL
1	12

```
SELECT country_id
FROM locations
GROUP BY country_id
HAVING COUNT(location_id) > 1;
```

	COUNTRY_ID
1	US

1.5

```
Code
SELECT department_id,
department_name,
AVG(salary)
FROM employees JOIN departments USING(department_id)
GROUP BY department_id, department_name
ORDER BY AVG(salary) ASC;
```

[illegible]

1.6

```
SELECT job_id, MAX(salary)
FROM employees
GROUP BY job_id
HAVING MAX(salary) > 10000;
```

	JOB_ID	MAX(SALARY)
1	AD_VP	17000
2	AC_MGR	12000
3	SA_MAN	10500
4	AD_PRES	24000
5	MK_MAN	13000
6	SA_REP	11000

1.7

Code

```
SELECT manager_id AS Manager,
COUNT(*) AS Mitarbeiteranzahl,
ROUND(AVG(salary), 2) AS Durchschnittsgehalt
FROM employees
WHERE manager_id IS NOT NULL
GROUP BY manager_id
HAVING COUNT(*) > 1;
```

Ausgabe

	MANAGER	MITARBEITERANZAHL	DURCHSCHNITTSGEHALT
1	124	4	2925
2	103	2	5100
3	101	2	8200
4	149	3	8866.67
5	100	5	12660

Aufgabe 2

2.1

Code

```
SELECT employee_id, last_name
FROM employees
WHERE salary > (
    SELECT AVG(salary)
    FROM employees
)
ORDER BY salary;
```

Ausgabe

	EMPLOYEE_ID	LAST_NAME
1	103	Hunold
2	149	Zlotkey
3	174	Abel
4	205	Higgins
5	201	Hartstein
6	101	Kochhar
7	102	De Haan
8	100	King

2.2a

Code

```
SELECT AVG(salary)
FROM employees
WHERE department_id = 10;
```

Ausgabe

	AVG(SALARY)
1	4400

2.2b

Code

-- b) Angestellte, die mehr als 4400 verdienen

```
SELECT employee_id, last_name
FROM employees
WHERE salary > 4400;
```

Ausgabe

	EMPLOYEE_ID	LAST_NAME
1	100	King
2	101	Kochhar
3	102	De Haan
4	103	Hunold
5	104	Ernst
6	124	Mourgos
7	149	Zlotkey
8	174	Abel
9	176	Taylor
10	178	Grant
11	201	Hartstein
12	202	Fay
13	205	Higgins
14	206	Gietz

2.2c

Code

-- c) Gemeinsam

```
SELECT employee_id, last_name, salary
FROM employees e
WHERE salary > (
    SELECT AVG(salary)
    FROM employees eSal
    WHERE e.department_id = eSal.department_id
)
ORDER BY salary; -- (Aufgabe 1 erweitern)
```

Ausgabe

	EMPLOYEE_ID	LAST_NAME
1	124	Mourgos
2	103	Hunold
3	149	Zlotkey
4	174	Abel
5	205	Higgins
6	201	Hartstein
7	100	King

2.3

Code

```

SELECT employee_id, last_name
FROM employees e
WHERE 2 < (
    -- Unterabfrage: Anzahl der Angestellten in der Abteilung
    SELECT COUNT(employee_id)
    FROM employees depEmp
    WHERE e.department_id = depEmp.department_id
);

```

Ausgabe

	EMPLOYEE_ID	LAST_NAME
1	100	King
2	101	Kochhar
3	102	De Haan
4	103	Hunold
5	104	Ernst
6	107	Lorentz
7	124	Mourgos
8	141	Rajs
9	142	Davies
10	143	Matos
11	144	Vargas
12	149	Zlotkey
13	174	Abel
14	176	Taylor

2.4

Code

```

SELECT department_name
FROM departments
WHERE department_id IN (
    -- Liste aller department_ids mit genau 2 mitarbeitern
    SELECT department_id
    FROM employees
    GROUP BY department_id
    HAVING COUNT(*) = 2
);

```

Ausgabe

	DEPARTMENT_NAME
1	Accounting
2	Marketing

2.5

Code

```

SELECT last_name, job_id, salary
FROM employees
WHERE salary > ALL (
    SELECT salary
    FROM employees
    WHERE job_id = 'IT_PROG'
)
ORDER BY salary DESC;

```

Ausgabe

	LAST_NAME	JOB_ID	SALARY
1	King	AD_PRES	24000
2	De Haan	AD_VP	17000
3	Kochhar	AD_VP	17000
4	Hartstein	MK_MAN	13000
5	Higgins	AC_MGR	12000
6	Abel	SA_REP	11000
7	Zlotkey	SA_MAN	10500

2.6

Code

```

SELECT last_name
FROM employees e
WHERE NOT EXISTS (
    SELECT manager_id
    FROM employees m
    WHERE e.employee_id = m.manager_id
);

```

Ausgabe

	LAST_NAME
1	Ernst
2	Lorentz
3	Rajs
4	Davies
5	Matos
6	Vargas
7	Abel
8	Taylor
9	Grant
10	Whalen
11	Fay
12	Gietz

2.7

Code

```

SELECT job_id, job_title, AVG(salary)
FROM employees JOIN jobs USING(job_id)
GROUP BY job_id, job_title
HAVING AVG(salary) >= ALL (
    -- Liste aller Durchschnittsgehälter
    SELECT AVG(salary)
    FROM employees
    GROUP BY job_id
);

```

Ausgabe

	JOB_ID	JOB_TITLE	AVG(SALARY)
1	AD_PRES	President	24000

2.8

Code

```

SELECT employee_id, last_name, job_id
FROM employees e
-- where job_history hat genau 2 einträge zu dem employee
WHERE 2 <= (
    SELECT COUNT(*)
    FROM job_history jh
    WHERE e.employee_id = jh.employee_id
);

```

Ausgabe

	EMPLOYEE_ID	LAST_NAME	JOB_ID
1	101	Kochhar	AD_VP
2	176	Taylor	SA_REP
3	200	Whalen	AD_ASST

2.9

Code

```

SELECT last_name, salary, department_id
FROM employees e
WHERE salary = (
    SELECT MIN(salary)
    FROM employees empDep
    WHERE empDep.department_id = e.department_id
);

```

Ausgabe

	LAST_NAME	SALARY	DEPARTMENT_ID
1	Kochhar	17000	90
2	De Haan	17000	90
3	Lorentz	4200	60
4	Vargas	2500	50
5	Taylor	8600	80
6	Whalen	4400	10
7	Fay	6000	20
8	Gietz	8300	110

2.10

Code

```

SELECT last_name, job_id, salary
FROM employees e
WHERE salary = (
    SELECT MIN(salary)
    FROM employees empDep
    WHERE empDep.department_id = e.department_id
) AND 1 <= (
    SELECT COUNT(*)
    FROM job_history jh
    WHERE e.employee_id = jh.employee_id
);

```

Ausgabe

	LAST_NAME	JOB_ID	SALARY
1	De Haan	AD_VP	17000
2	Taylor	SA_REP	8600
3	Kochhar	AD_VP	17000
4	Whalen	AD_ASST	4400

2.11

Code

```

SELECT first_name, last_name, e.salary, manager_id
FROM employees e
ORDER BY (
  SELECT m.salary - e.salary
  FROM employees m
  WHERE m.employee_id = e.manager_id
) DESC;

```

Ausgabe

	FIRST_NAME	LAST_NAME	SALARY	MANAGER_ID
1	Steven	King	24000	(null)
2	Kevin	Mourgos	5800	100
3	Eleni	Zlotkey	10500	100
4	Jennifer	Whalen	4400	101
5	Michael	Hartstein	13000	100
6	Alexander	Hunold	9000	102
7	Lex	De Haan	17000	100
8	Neena	Kochhar	17000	100
9	Pat	Fay	6000	201
10	Shelley	Higgins	12000	101
11	Diana	Lorentz	4200	103
12	William	Gietz	8300	205
13	Kimberely	Grant	7000	149
14	Peter	Vargas	2500	124
15	Randall	Matos	2600	124
16	Bruce	Ernst	6000	103
17	Curtis	Davies	3100	124
18	Trenna	Rajs	3500	124
19	Jonathon	Taylor	8600	149
20	Ellen	Abel	11000	149

Aufgabe 3

3.1

Code

```
SELECT last_name
FROM employees
WHERE employee_id NOT IN (
    SELECT DISTINCT manager_id
    FROM employees
    WHERE manager_id IS NOT NULL
);
```

Ausgabe

	LAST_NAME
1	Ernst
2	Lorentz
3	Rajs
4	Davies
5	Matos
6	Vargas
7	Abel
8	Taylor
9	Grant
10	Whalen
11	Fay
12	Gietz

3.2

Code

```
SELECT e.last_name,
e.job_id,
e.salary
FROM employees e JOIN
(
    SELECT department_id, MIN(salary) AS min_sal
    FROM employees
    GROUP BY department_id
) dep_min ON e.department_id = dep_min.department_id AND
e.salary = dep_min.min_sal
JOIN (
    SELECT employee_id
    FROM job_history
    GROUP BY employee_id
) jChange ON jChange.employee_id = e.employee_id;
```

Ausgabe

	LAST_NAME	JOB_ID	SALARY
1	Kochhar	AD_VP	17000
2	De Haan	AD_VP	17000
3	Whalen	AD_ASST	4400
4	Taylor	SA_REP	8600